

Spatial Data Analysis for a Spatial World

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Where can I find these slides?

Raquel Menezes's GitHub

<http://github.com/rmenezesmotaite/SummerSchoolUV>



① Introduction to Spatial Statistics

- ▷ Why Does Location Matter?
- ▷ Data Analysis and Visualization

② Three Types of Spatial Data

- ▷ Try a QUIZ

③ A Closer Look at Geostatistics

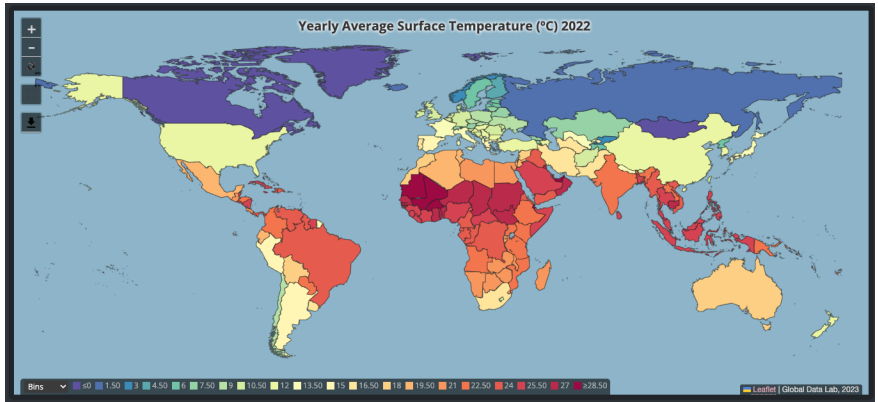
- ▷ Spatial Correlation

④ Fisheries and Marine Applications

- ▷ How Can Spatial Statistics Help?
- ▷ Try DYNAMIC MAPS

Where have you seen spatial data in real life?

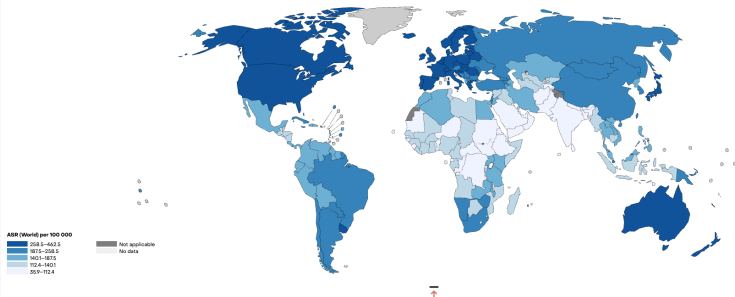
Climate Data



Global Data Lab: [link](#)

Disease Data

Age-Standardized Rate (World) per 100 000, Incidence, Both sexes, in 2022
All cancers



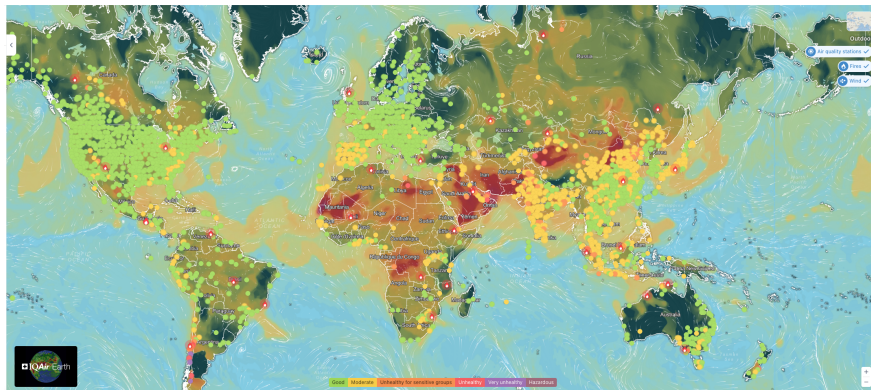
All rights reserved. The designations employed and the presentation of the material in this publication do not imply the expression of any opinion whatsoever on the part of the World Health Organization / International Agency for Research on Cancer concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate borderlines for which there may not yet be full agreement.

Cancer TODAY | IARC
https://gobcan.iarc.fr/home
Data version: Globocan 2022 (version 1) - 08.02.2024
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International Agency
for Research on Cancer
World Health
Organization

World Health Organization: [link](#)

Pollution Data



IQAir: [link](#)

An Early Landmark in Spatial Data Analysis

1854 London Cholera Outbreak

Background

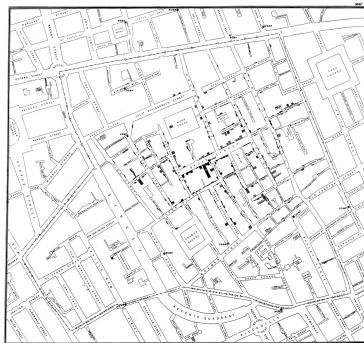
- **John Snow (1813–1858):** British physician and one of the pioneers of epidemiology.

The Outbreak

- **London, 1854:** A severe cholera outbreak struck Soho.
- **Common belief:** Cholera spread through “bad air”.

Snow's Insight

- Cholera was **waterborne**, not airborne.
- He **mapped locations of cholera deaths**.



What Did the Map Reveal?

- Most cholera deaths clustered around the **Broad Street water pump**.
- The pump handle was removed, and the outbreak quickly declined.



Key Message

Spatial patterns revealed the source of the epidemic.

Exploratory Data Analysis (EDA)

Before building any statistical model, we first need to **understand the data**.

EDA (also called [data mining](#)) helps us answer questions such as:

- What do the data look like?
- Are there unusual observations or outliers?
- Are there relationships between variables?
- Are there spatial patterns?

Key Message

First explore. Then model.

How Do We Explore the Data?

EDA relies on two complementary approaches:

Numbers

- mean
- median
- quartiles
- variance

Pictures

- histograms
- boxplots
- scatterplots
- spatial maps

Important

Both are essential.

Why Summary Statistics Are Not Sufficient?

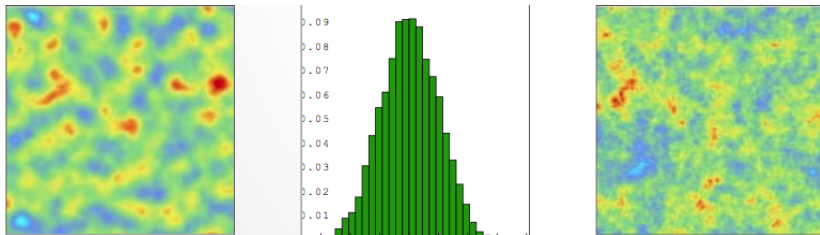


Figure: Two different spatial maps (**highest values** and **lowest values**) with same histogram and same numerical summaries (mean, standard deviation, etc)

Key Message

Numbers alone can be misleading.

Visualisation helps us understand the data.

Different plots answer different questions:

- Histograms show distributions.
- Scatterplots show relationships.
- **Maps reveal spatial patterns.**
- Time series show change over time.

Key Message

Maps are one of the most powerful tools in spatial data analysis.

Example: Meuse River Data (*gstat* R package)

Study area

Heavy metal concentrations (ppm) were measured in topsoil at sampling locations in the Meuse River floodplain (Netherlands).

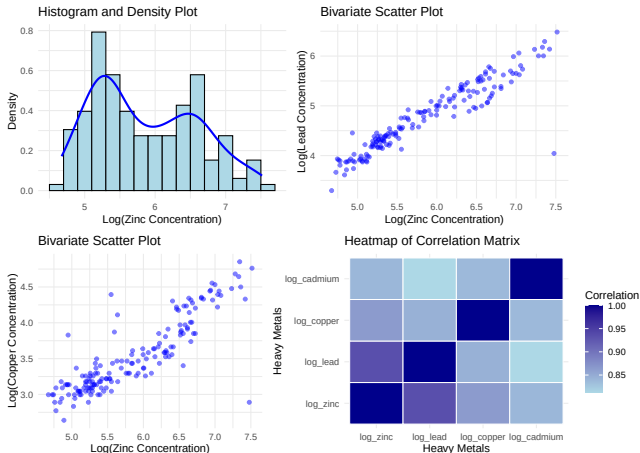
Variables

- Cadmium
- Copper
- Lead
- Zinc

This dataset will illustrate some common exploratory spatial data analysis techniques.



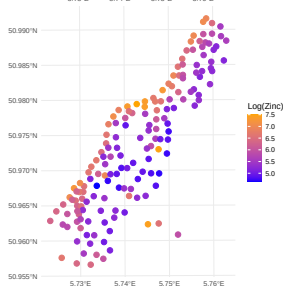
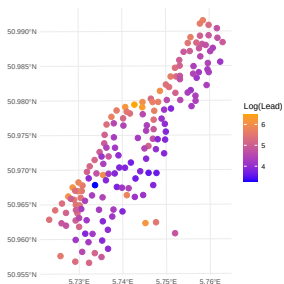
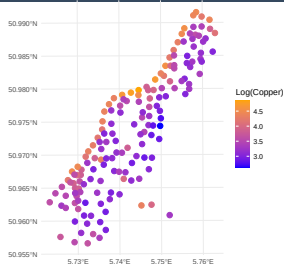
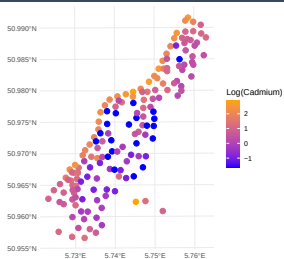
Exploring the Variables



Key Message

Different visualisations reveal different aspects of the data.

Visualising Spatial Patterns



Key Message

Maps reveal how measurements vary across space.

- ① Introduction to Spatial Statistics
 - ▷ Why Does Location Matter?
 - ▷ Data Analysis and Visualization

- ② Three Types of Spatial Data
 - ▷ Try a QUIZ

- ③ A Closer Look at Geostatistics
 - ▷ Spatial Correlation

- ④ Fisheries and Marine Applications
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Three Types of Spatial Data

Although all spatial data have a geographic location, they are **not all of the same type**.

In spatial statistics, we usually distinguish three main types:

① **Areal data**

Observations aggregated over regions or polygons

② **Spatial point patterns**

Locations of events or objects

③ **Geostatistical data**

Measurements collected at sampling locations

Each type requires different statistical methods.

1. Areal Data

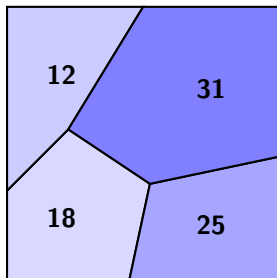
Observations are **aggregated over predefined regions** rather than recorded at exact locations.

Typical examples include:

- Disease incidence by municipality
- Unemployment rate by district
- Fish catches by ICES statistical rectangle

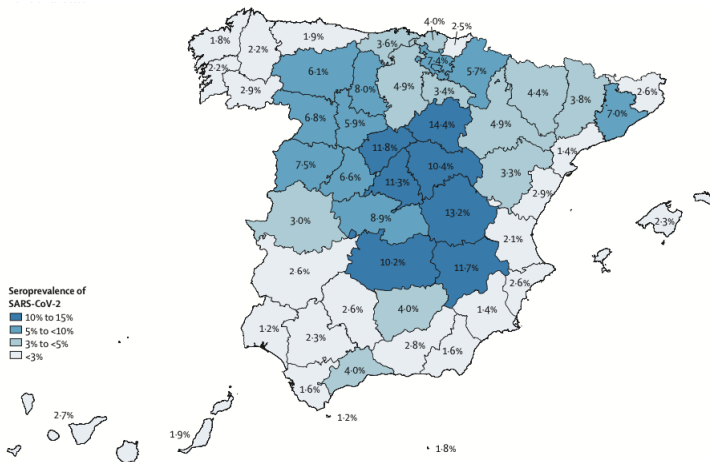
Key Question

Do neighbouring regions tend to have similar values?



One value per region

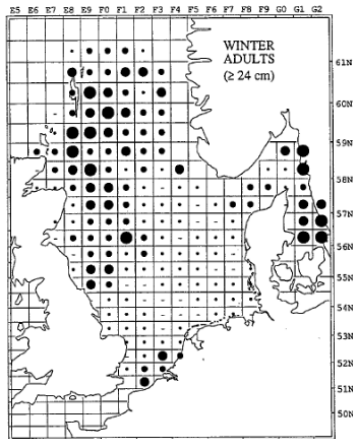
Example: COVID-19 Prevalence in Spain



Prevalence of SARS-CoV-2 in Spain, serological survey from April 27 to May 11, 2020, involving 61,075 participants (Pollán et al., 2020)

Example: Fisheries Data

Fish catches are often aggregated within ICES statistical rectangles. Each rectangle contains one value, such as the estimated abundance or catch of a species.



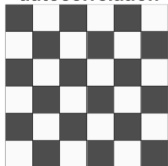
Each ICES rectangle represents one observation.

Can each region be analysed independently?

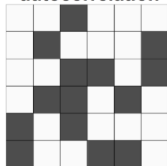
Usually, the answer is **NO**.

Nearby regions often share similar environmental, social or demographic characteristics.

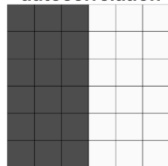
Negative spatial autocorrelation



No spatial autocorrelation



Positive spatial autocorrelation



Most Common

Why does it matter?

Spatial statistical models (e.g. CAR and BYM models) explicitly account for this dependence between neighbouring regions.

2. Spatial Point Patterns

The **locations themselves** are the data, rather than values aggregated over regions.

Examples include:

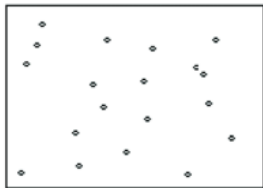
- Locations of disease cases
- Locations of trees in a forest
- Locations of dolphin sightings
- Locations of fishing vessels

Key Idea

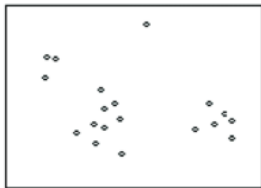
One point represents one event (or one object).

Spatial Point Patterns

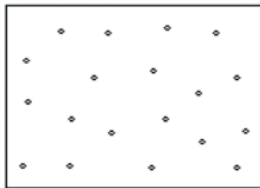
Random



Aggregated



Regular



Think about it:

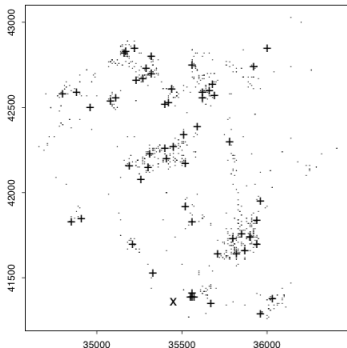
- Which pattern would you expect if trees compete for space?
- Which pattern would you expect during a disease outbreak?

Key Message

Point pattern analysis helps us determine whether events are **random, clustered, or regularly spaced**.

Example: Lung and Larynx Cancers

Each point represents the location of one patient, rather than an aggregated value for a region.



Legend:

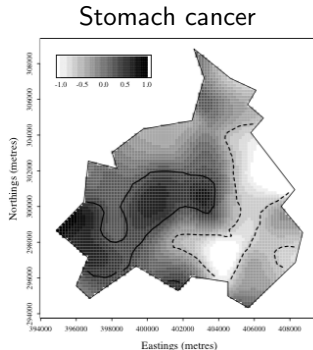
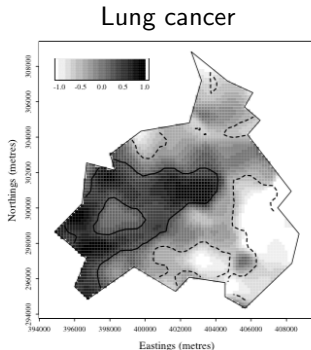
- cases of lung cancer (dots)
- cases of larynx cancer (crosses)
- a deactivated industrial incinerator (x symbol)

Questions we may ask:

- Do the cases cluster?
- Is disease risk higher near the incinerator?
- Does disease risk vary across space?

From Events to Risk Maps

Observed disease locations can be used to estimate continuous maps of disease risk.



- Both diseases show similar spatial patterns.
- Contours highlight areas with higher or lower estimated risk.

3. Geostatistical Data

Measurements are collected at **sampling locations** within a continuous spatial region.

Examples include:

- Air pollution
- Sea surface temperature
- Salinity
- Soil contamination
- Fish abundance

Key Idea

One measurement per sampling location.

The Main Challenge

The phenomenon of interest exists **everywhere in space**,
but measurements are available only at a limited number of locations.

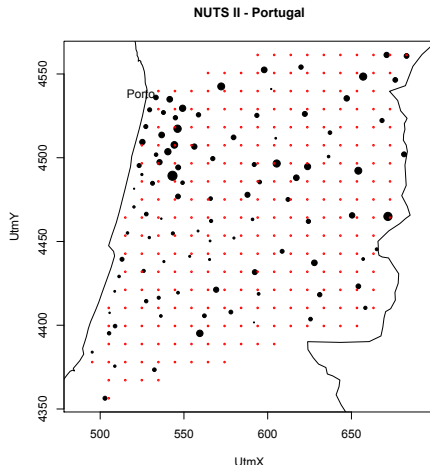
Key Question

How can we estimate values at unsampled locations?

Spatial interpolation methods, such as **kriging**, provide the answer.

Example: Soil Contamination

Measurements of arsenic concentration are available at a limited number of sampling locations.

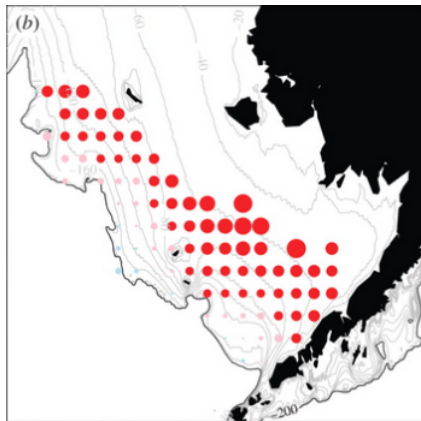


Objective

Predict arsenic concentration at locations where no measurements are available.

Example: Fish Abundance in the Bering Sea

Measurements of adult fish abundance are collected at sampling locations.



Goal

Estimate fish abundance throughout the study region.

Three Types of Spatial Data

Areal data	One value per region.
Point patterns	One point per event.
Geostatistical data	One measurement per sampling location.

Remember

Choosing the appropriate statistical method starts with identifying the type of spatial data.

Interactive Quiz – Which type of spatial data is most appropriate?

Areal data? Spatial point patterns? Geostatistical data?

Instructions:

- Use your phone to scan the QR code
- Take the quiz to match each case study with the corresponding area of spatial statistics



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Fundamental Concept of “Spatial Dependence”

Everything is related to everything else, but near things are more related than distant things (first law of geography)

Waldo Tobler, 1970

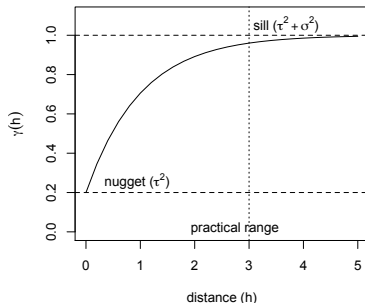
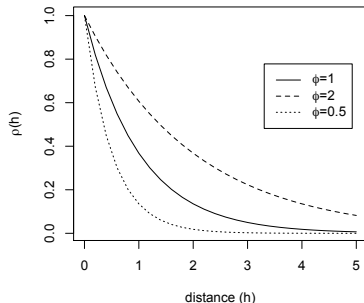
Key Idea

Spatial correlation usually decreases with distance.

Beyond a certain distance, observations often become approximately independent.

Describing Spatial Correlation

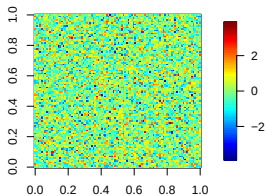
Spatial dependence can be described in different ways.



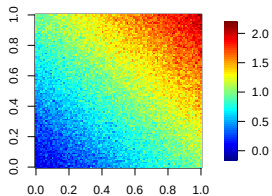
- **Correlation function:** shows how correlation decreases with distance.
- **Variogram:** measures how dissimilar observations become as distance increases.

Simulated Spatial Surfaces

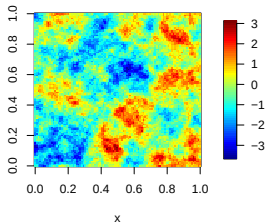
White Noise



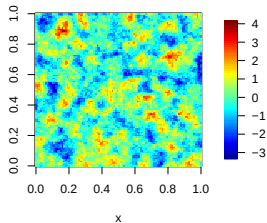
Non-stationary RF



Stationary RF (exp)



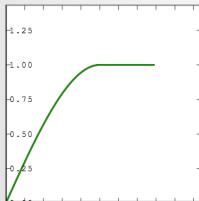
Stationary RF (sph)



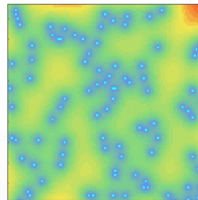
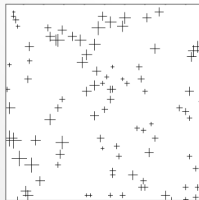
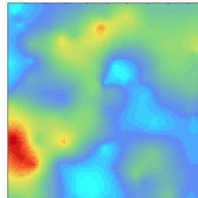
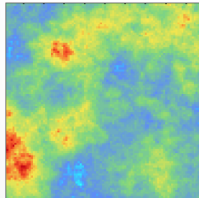
From Measurements to Prediction

- I Simulate a spatial surface.
- II Sample 100 observation locations.
- III Predict the complete surface using kriging.
- IV Compare the prediction with the true surface.

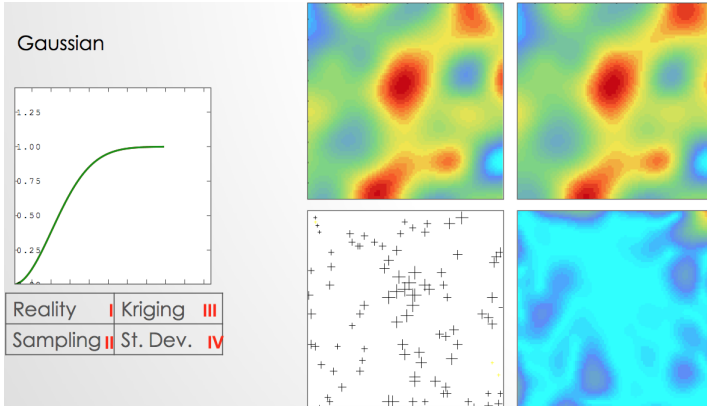
Spherical



Reality	I	Kriging	III
Sampling	II	St. Dev.	IV



From Measurements to Prediction



Key Message

Spatial correlation allows us to predict values at locations where no measurements are available.

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Marine Spatial Planning (MSP)

Marine Spatial Planning aims to balance **human activities** while protecting **marine ecosystems**.

- Fisheries
- Shipping
- Offshore energy
- Aquaculture
- Tourism
- Marine conservation

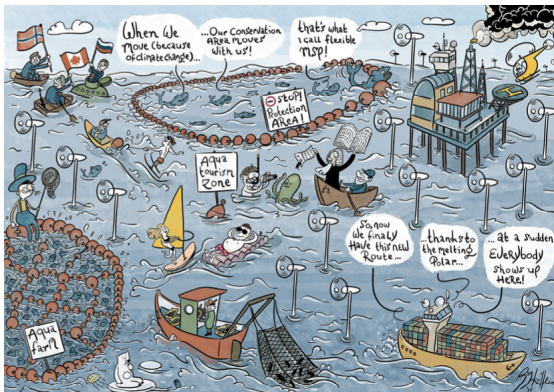


Figure: A crowded ocean under a changing climate (Santos et al., 2020).

How Can Spatial Statistics Help?

Spatial data analysis supports marine management by helping us

- map species distributions;
- predict fish abundance;
- identify biodiversity hotspots;
- monitor environmental change;
- support sustainable fisheries.

Key Message

Spatial data analysis transforms observations into information for evidence-based decision making.

Where Do the Data Come From?

Scientific surveys

- Representative sampling
- Expensive
- Limited in time

Commercial fisheries

- Large amounts of data
- Lower cost
- Preferential sampling

Satellite data

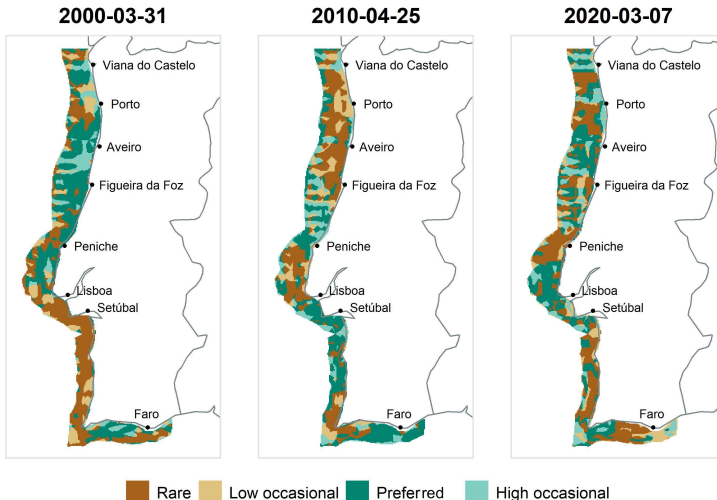
- Sea surface temperature
- Chlorophyll
- Ocean currents

Key Message

Different data sources provide complementary information.

Spatial Statistics Supporting Decision Making

Example: Sardine Occupancy Areas



Key Message

Spatial prediction maps support sustainable fisheries management and

Take-Home Messages

- ① Spatial data are everywhere.
- ② Different spatial data require different statistical methods.
- ③ Spatial statistics transforms data into knowledge for better decision making.

Thank you!

Now It's Your Turn!

Let's explore some interactive spatial data visualizations.

Instructions

- ① Scan the QR code.
- ② Open the interactive webpage.
- ③ Explore the maps and graphics.
- ④ Zoom in to the Portuguese coast.

